

Computer Programming Assignment No:2

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Section B

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Submitted to:

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Course:

EE170 Computer Programming



- Take 10 integer inputs in an array.
- Find and display: Maximum value Minimum value Average

```
#include <iostream>
using namespace std;

int main() {
    int arr[10];
    int max, min;
    int sum = 0;
    cout << "Enter 10 numbers:\n";
    for(int i = 0; i < 10; i++) {
        cin >> arr[i];
        sum += arr[i];
    }
    max = min = arr[0];

    for(int i = 1; i < 10; i++) {
        if(arr[i] > max)
            max = arr[i];
        if(arr[i] < min)
            min = arr[i];
    }

    cout << "Maximum: " << max << endl;
    cout << "Minimum: " << min << endl;
    cout << "Average: " << sum / 10.0 << endl;

    return 0;
}
```

```
// For input/output
// To use cout and cin without std::
// Main function starts
// Declare array of size 10
// Variables to store maximum and minimum
// Variable to store sum (initialized to 0)
// Ask user for input
// Loop to take 10 inputs
// Store input in array
// Add value to sum
// Assume first element is max and min
// Loop from second element
// Check for maximum
// Update max
// Check for minimum
// Update min
// Print max
// Print min
// Print average
// End program
```

2. Write a program that:

- **Uses a function to reverse a 1D array.**
- **Display the array before and after reversal.**

```
#include <iostream>
```

```
// Library for input/output
```

```
using namespace std;
```

```
// Standard namespace
```

```
// Function to reverse array
```

```
void reverseArr(int arr[], int size) {
```

// Function with array and size

```
for(int i = 0; i < size / 2; i++) {
```

```
// Loop till half array
```

```
int temp = arr[i];
```

```
// Store first element
```

```
arr[i] = arr[size - 1 - i];
```

```
// Replace with last element
```

```
arr[size - 1 - i] = temp;
```

```
// Put stored value at end
```

}

}

```
int main() {
```

```
int arr[5];
```

```
// Declare array of size 5
```

```
cout << "Enter 5 elements:\n";
```

```
// Ask for input
```

```
for(int i = 0; i < 5; i++)
```

```
// Loop for input
```

```
cin >> arr[i];
```

```
// Store value
```

```
cout << "Before: ";
```

```
// Print before reversing
```

```
for(int i = 0; i < 5; i++)
```

```
cout << arr[i] << " ";
```

```
reverseArr(arr, 5);
```

```
// Call function
```

```
cout << "\nAfter: ";
```

```
// Print after reversing
```

```
for(int i = 0; i < 5; i++)
```

```
cout << arr[i] << " ";
```

```
return 0;
```

```
// End program
```

}

4. Write a program that:

- Stores names of 5 students in a string array.
- Sorts them in alphabetical order.
- Displays the sorted list.

```
#include <iostream>                                // Input/output
using namespace std;
int main() {
    string names[5];                                // Array of 5 strings
    cout << "Enter 5 names:\n";                    // Input names
    for(int i = 0; i < 5; i++)
        cin >> names[i];
    for(int i = 0; i < 5; i++)                        // Outer loop
        for(int j = i + 1; j < 5; j++)              // Inner loop
            if(names[i] > names[j])                 // Compare strings
                swap(names[i], names[j]);           // Swa
    cout << "Sorted Names:\n";                      // Display result
    for(int i = 0; i < 5; i++)
        cout << names[i] << endl;
    return 0;
}
```

5. Power System Load Analysis (1D Array + Function) In a power system, hourly load demand (in MW) is recorded. Write a program to:

- Store load data for 24 hours in an array.
- Create functions to calculate:

Total load

Peak load

```
#include <iostream>
using namespace std;
int totalLoad(int arr[]) { // Function to calculate total
    int sum = 0;
    for(int i = 0; i < 24; i++) // Loop through 24 hours
        sum += arr[i]; // Add values
    return sum; // Return total
}
int peakLoad(int arr[]) { // Function for max value
    int max = arr[0];
    for(int i = 1; i < 24; i++)
        if(arr[i] > max)
            max = arr[i];
    return max;
}
int main() {
    int load[24]; // Array for 24 hours
    cout << "Enter 24 load values:\n";
    for(int i = 0; i < 24; i++)
        cin >> load[i];
    cout << "Total Load: " << totalLoad(load) << endl;
    cout << "Peak Load: " << peakLoad(load) << endl;
    return 0;}
```

6. Design a program to: Store student names (string array) and their marks (1D array).

Use functions to:

- **Find topper**
- **Display pass/fail status**
- **Search a student by name**

```
#include <iostream>
```

```
using namespace std;
```

```
void topper(string name[], int marks[]) { // Find topper
```

```
    int max = marks[0], index = 0;
```

```
    for(int i = 1; i < 5; i++)
```

```
        if(marks[i] > max) {
```

```
            max = marks[i];
```

```
            index = i;}
    cout << "Topper: " << name[index] << endl;}
```

```
void passFail(string name[], int marks[]) { // Pass/Fail
```

```
    for(int i = 0; i < 5; i++)
```

```
        cout << name[i] << " "
```

```
        << (marks[i] >= 50 ? "Pass" : "Fail") << endl;}
```

```
void search(string name[], int marks[], string key) { // Search
```

```
    for(int i = 0; i < 5; i++)
```

```
        if(name[i] == key)
```

```
            cout << "Marks: " << marks[i] << endl;}
```

```
int main() {
```

```
    string name[5];    int marks[5];
```

```
    for(int i = 0; i < 5; i++)
```

```
        cin >> name[i] >> marks[i];
```

```
    topper(name, marks);
```

```
    passFail(name, marks);
```

```
    search(name, marks, "Ali");    return 0;}
```

7. Renewable Energy Monitoring System (1D Array + Functions) A solar power plant records the energy generated (in kWh) every hour for one day. Write a C++ program to:

Store energy generation data for 24 hours in a 1D array.

Create functions to:

- Calculate total energy generated in a day
- Find the hour with maximum generation
- Calculate average energy generation
- Display all results clearly.

```
#include <iostream>

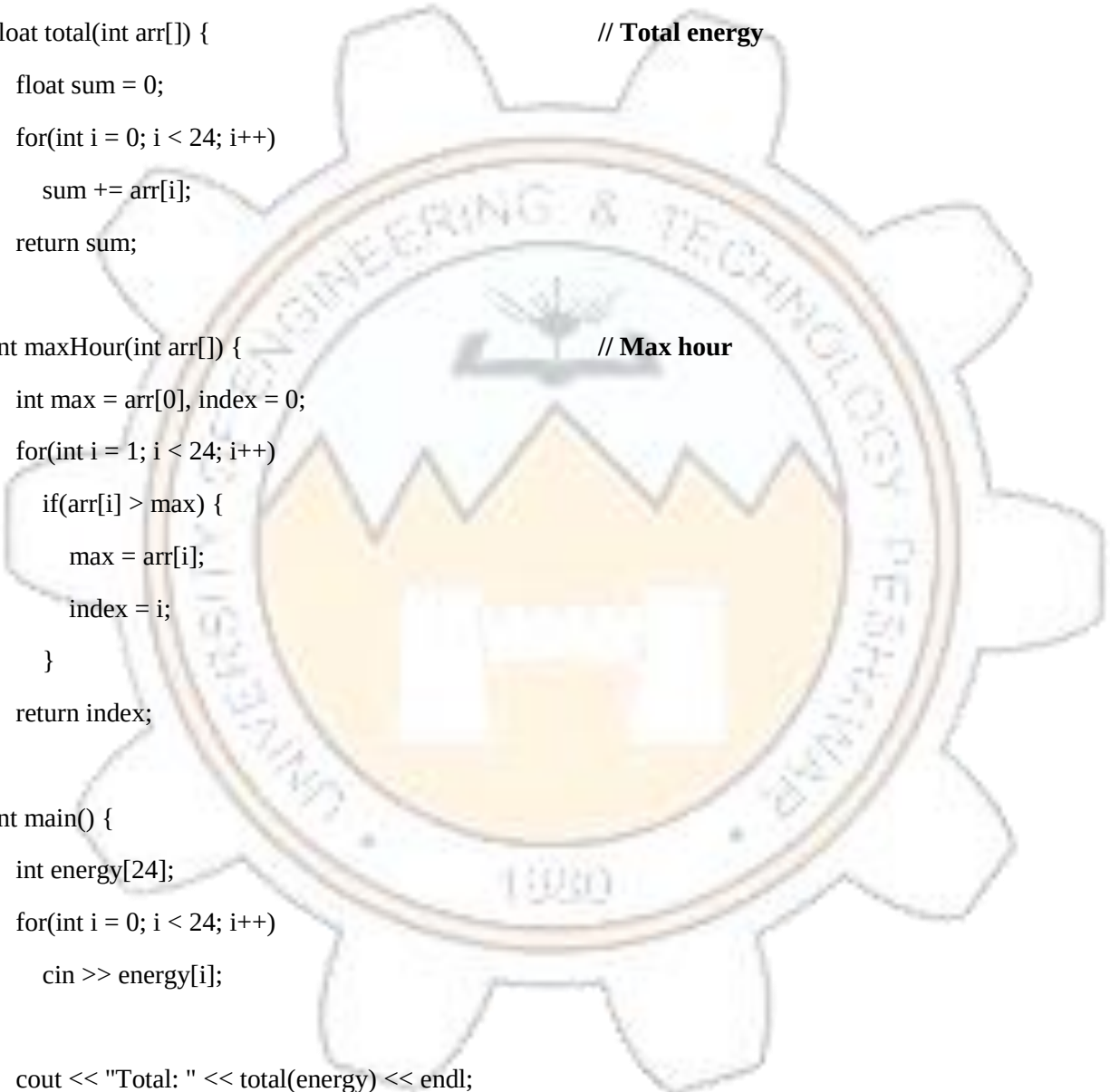
using namespace std;

float total(int arr[]) { // Total energy
    float sum = 0;
    for(int i = 0; i < 24; i++)
        sum += arr[i];
    return sum;
}

int maxHour(int arr[]) { // Max hour
    int max = arr[0], index = 0;
    for(int i = 1; i < 24; i++)
        if(arr[i] > max) {
            max = arr[i];
            index = i;
        }
    return index;
}

int main() {
    int energy[24];
    for(int i = 0; i < 24; i++)
        cin >> energy[i];

    cout << "Total: " << total(energy) << endl;
    cout << "Max Hour: " << maxHour(energy) << endl;
    cout << "Average: " << total(energy)/24 << endl;
    return 0;
}
```



8. Battery Health Monitoring (1D Array + Functions) In battery management systems, voltage readings are monitored to ensure safe operation. Write a program to:

- Store 20 voltage readings of a battery in an array.
- Define safe voltage limits (e.g., 3.0V to 4.2V).
- Create functions to:
- Count how many readings are below or above safe limits
- Find the maximum and minimum voltage
- Display a warning message if unsafe readings exist

```
#include <iostream>
using namespace std;
int main() {
    float v[20];                // Voltage array
    int low = 0, high = 0;      // Counters
    float max, min;
    for(int i = 0; i < 20; i++) {
        cin >> v[i];
        if(i == 0) {           // Initialize max and min
            max = min = v[i];
        }
        if(v[i] < 3.0) low++;    // Below safe limit
        if(v[i] > 4.2) high++;   // Above safe limit
        if(v[i] > max) max = v[i];
        if(v[i] < min) min = v[i];
    }
    cout << "Low: " << low << " High: " << high << endl;
    cout << "Max: " << max << " Min: " << min << endl;
    if(low > 0 || high > 0)
        cout << "Warning: Unsafe readings!\n";
}
```

```
    return 0;
}
```

9. Smart Grid Load Distribution (2D Array + Functions) In a smart grid, electricity consumption is recorded across multiple regions. Write a program to:

- Store power consumption (in MW) for 4 regions over 7 days using a 2D array.
- Create functions to:
- Calculate total consumption for each region
- Identify the day with highest total demand
- Compute overall average consumption

```
#include <iostream>
using namespace std;

int main() {
    int grid[4][7]; // 4 regions, 7 days
    for(int i = 0; i < 4; i++) // Input data
        for(int j = 0; j < 7; j++)
            cin >> grid[i][j];
    for(int i = 0; i < 4; i++) { // Total per region
        int sum = 0;
        for(int j = 0; j < 7; j++)
            sum += grid[i][j];
        cout << "Region " << i+1 << ": " << sum << endl;
    }
    int maxDay = 0, maxSum = 0;

    for(int j = 0; j < 7; j++) { // Check each day
        int sum = 0;
        for(int i = 0; i < 4; i++)
            sum += grid[i][j];
        if(sum > maxSum) {
            maxSum = sum;
            maxDay = j;
        }
    }
}
```

```

    }
}

cout << "Highest demand day: " << maxDay + 1 << endl;

int total = 0;

for(int i = 0; i < 4; i++)
    for(int j = 0; j < 7; j++)
        total += grid[i][j];

cout << "Average: " << total / 28.0 << endl;

return 0;
}

```

10. Classroom Attendance System (2D Array + Functions) Teachers often track student attendance daily. Write a program to:

- Store attendance for 5 students over 7 days using a 2D array (Use 1 for present, 0 for absent).
- Create functions to:
- Calculate total attendance of each student
- Find the student with highest attendance
- Calculate overall class attendance percentage

```

#include <iostream>                                // Library for input and output
using namespace std;                              // Use standard namespace

// Function to calculate total attendance of each student

void totalAttendance(int att[][7])                // 2D array with 7 columns (days)
{
    for(int i = 0; i < 5; i++)                     // Loop through 5 students (rows)
    {
        int total = 0;                             // Initialize total for each student
        for(int j = 0; j < 7; j++)                 // Loop through 7 days (columns)

```

```

{
    total += att[i][j];
}

// Display total attendance of each student
cout << "Student " << i + 1
    << " total attendance: " << total << endl;
}
}

// Function to find student with highest attendance
void highestAttendance(int att[][7])
{
    int max = -1;
    attendance // Variable to store maximum
    int student = -1; // Variable to store student index
    for(int i = 0; i < 5; i++) // Loop through students
    {
        int total = 0; // Reset total for each student
        for(int j = 0; j < 7; j++) // Loop through days
        {
            total += att[i][j]; // Calculate total attendance
        }
        if(total > max) // Check if current total is greater
        {
            max = total; // Update maximum
            student = i; // Store student index
        }
    }

// Display result
cout << "Student with highest attendance: Student "
    << student + 1 << " (" << max << " days)" << endl;
}

```

// Function to calculate overall class attendance percentage

```
void overallAttendance(int att[][7])
```

```
{
```

```
    int total = 0;
```

// Variable to store total attendance

```
    for(int i = 0; i < 5; i++)
```

// Loop through students

```
    {
```

```
        for(int j = 0; j < 7; j++)
```

// Loop through days

```
        {
```

```
            total += att[i][j];
```

// Add all attendance values

```
        }
```

```
    }
```

// Total possible attendance = students × days = 5 × 7 = 35

```
    float percentage = (total / 35.0) * 100;
```

// Calculate percentage

```
    cout << "Overall class attendance: "
```

```
        << percentage << "%" << endl;
```

```
}
```

```
int main()
```

// Main function starts

```
{
```

```
    int att[5][7];
```

// 2D array for 5 students & 7 days

```
    cout << "Enter attendance (0 = Absent, 1 = Present):\n";
```

// Input attendance data

```
    for(int i = 0; i < 5; i++)
```

// Loop for students

```
    {
```

```
        for(int j = 0; j < 7; j++)
```

// Loop for days

```
        {
```

```
            cin >> att[i][j];
```

// Take input

// Validate input (must be 0 or 1)


```
if(att[i][j] != 0 && att[i][j] != 1)
{
    cout << "Invalid input! Enter 0 or 1 again:\n";
    j--; // Repeat same position
}
}
cout << "\n--- Results ---\n";

totalAttendance(att); // Call function for each student total
highestAttendance(att); // Call function for highest attendance
overallAttendance(att); // Call function for overall percentage
return 0; // End program
}
```

